

J. BRANDON CARTER

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EDUCATION

The University of Texas at Austin

Austin, TX

PhD in Statistics, December 2024

Advisor: Catherine Calder

Dissertation: Bayesian Spatial Models for Discrete Data: Methodological Advances with Applications in Urban Sociology

Brigham Young University

Provo, UT

MS in Statistics, April 2019

Advisor: David B. Dahl

BS in Statistics, Magna Cum Laude, April 2019

Minors: Mathematics, World Dance

Speaker at College Graduation

RESEARCH INTERESTS

Methodological

Spatial and Spatio-temporal Statistics, Probabilistic Graphical Models, Bayesian Modeling and Computation

Applied

Social Science, Environmental Science, Assortment Planning

EXPERIENCE

The Hershey Company

Dallas, TX

Data Scientist, March 2026 - Present

Duke University

Durham, NC

Postdoctoral Scholar, June 2025 - February 2026

The University of Texas at Austin

Austin, TX

Postdoctoral Researcher, January 2025 - May 2025

Graduate Research Assistant, August 2020 - December 2024

Assistant Instructor, January 2023 - May 2023

Teaching Assistant, August 2022 - December 2022

Brigham Young University

Provo, UT

Research Assistant, August 2016 - April 2019

Teaching Assistant, August 2015 - April 2017

AWARDS AND HONORS

Savage Award: Applied Methodolgy, Finalist, 2026

ISBA Travel Award, 2024, 2026

Keller Award for leadership, service and community building, Department of Statistics and Data Sciences, The University of Texas at Austin, 2024

Professional Development Travel Award, College of Natural Sciences, The University of Texas at Austin, 2023

SDS Excellence Fellowship, 2 year, Department of Statistics and Data Sciences, The University of Texas at Austin

Academic scholarship, full tuition, Brigham Young University, August 2015 - April 2019

Mary B. Jensen Scholarship, full tuition, Brigham Young University, August 2018 - April 2019

Viltis Scholarship, half tuition, Brigham Young University, August 2017 - April 2018

PUBLICATIONS

Refereed Publications

J. B. Carter and C. A. Calder (2026). Mixture of directed graphical models for discrete spatial random fields. *Journal of the American Statistical Association*. Forthcoming [arXiv]

N. Pinchak, C. R. Browning, B. Boettner, **J. B. Carter**, and C. A. Calder (2026). Defended neighborhoods, vulnerable routines: routine activities, collective efficacy, and the social ecology of discrimination. *Social Forces*. Forthcoming

C. A. Calder, N. Pinchak, C. R. Browning, **J. B. Carter**, E. Hsiao, B. Boettner, J. Tarrence, and P. Bellair (2025). Mobility network-based measurement of local collective efficacy and its consequences for the spatial patterning of violent crime. *Journal of Quantitative Criminology* [link]

J. B. Carter, C. R. Browning, B. Boettner, N. Pinchak, and C. A. Calder (2024). Land-use filtering for nonstationary spatial prediction of collective efficacy in an urban environment. *The Annals of Applied Statistics* 18.1, pp. 794–818 [link]

D. B. Dahl, J. Andros, and **J. B. Carter** (2023). Cluster analysis via random partition distributions. *Statistical Analysis and Data Mining: The ASA Data Science Journal* 16.2, pp. 135–148 [link]

PRESENTATIONS

Invited Talks

Mixtures of Directed Graphical Models: A Tractable Alternative to Markov Random Fields for Bayesian Spatial Analysis

ISBA World Meeting, Nagoya, JP, July 2026

Data Jamboree: Web Scraping in Julia

ASA Computing Section Mini-Symposium, Online, November 2024

Land-use Filtering for Nonstationary Spatial Prediction of Collective Efficacy in an Urban Environment
CMStatistics, London, UK, December 2022

Invited Posters

Mixture of Directed Graphical Models for Discrete Spatial Random Fields

Joint Statistical Meetings, Nashville, TN, August 2025

Posters/Contributed Talks

Mixture of Directed Graphical Models for Discrete Spatial Random Fields
Joint Statistical Meetings, Portland, OR, August 2024

Mixture of Directed Graphical Models for Discrete Spatial Random Fields
ISBA World Meeting, Venice, IT, July 2024

Land-use Filtering for Nonstationary Spatial Prediction of Collective Efficacy in an Urban Environment
Joint Statistical Meetings, Toronto, CA, August 2023

Land-use Filtering for Nonstationary Spatial Prediction of Collective Efficacy in an Urban Environment
ENVR Workshop, Provo, UT, October 2022

FUNDING

NSF 2502935 (Role: PI)
Postdoctoral Fellowship: MSPRF: Next Generation Spatial Models for Discrete-valued, High-dimensional Data
Funding Period: 07/01/2025 - 07/31/2027, Total Award: \$190,000

TEACHING

The University of Texas at Austin Austin, TX
Assistant Instructor, SDS 320E Elements of Statistics, Spring 2023
Teaching Assistant, SDS 322E Elements of Data Science, Fall 2022

UNIVERSITY SERVICE

The University of Texas at Austin Austin, TX
Dean's Office Graduate Council, January 2021 - December 2024
Department Representative, Graduate Student Assembly, August 2020 - May 2022

PROFESSIONAL SERVICE

Organized Sessions

Scalable and Interpretable: Advances in Graphical Techniques for Modeling Spatial Dependence
ISBA World Meeting, Nagoya, JP, July 2026